Mengqi LI

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Linkedin | Scholar | Github

Ph.D. Student in Geography

Research Interests

- GeoAI
- Human Mobility
- Disaster Resilience

Education

- **University at Buffalo, SUNY** *Aug. 2025 - Present*
  Ph.D. in Geography (Specialization in GeoAI & Disaster Resilience) Buffalo, NY
  - **Advisor:** Prof. Yingjie Hu
- **University of Zurich** *Sep. 2023 - Jun. 2025*
  M.S. in Geographic Information Science and Systems Zurich, Switzerland
  - **Advisor:** Prof. Cheng Fu, Prof. Robert Weibel
- **Nanjing University of Information Science & Technology** *Sep. 2019 - Jun. 2023*
  B.S. in Geographic Information Science Nanjing, China

Peer-Reviewed Publication

[1] **Li, M.**, Weibel, R., Hu, Y., Fu, C., "Understanding the Resilience of Different Urban Mobility Systems to Extreme Weather Events: A Case Study of Hurricane Ida". 2026

[2] **Li, M.**, Dai, W., Wang, G., Wang, B., Chen, K., Gao, Y., Amankwah, S. O. Y., "Reconstructing high-resolution DEMs from 3D terrain features using conditional generative adversarial networks". In: *International Journal of Applied Earth Observation and Geoinformation* 133 (2024), p. 104115

[3] Luo, W., Liu, Z., Ran, Y., **Li, M.**, Zhou, Y., Hou, W., Lai, S., "Unraveling varying spatiotemporal patterns of Dengue Fever and associated exposure-response relationships with environmental variables in three Southeast Asian countries before and during COVID-19". In: *PLOS Neglected Tropical Diseases* 19.4 (2025), e0012096

[4] Wang, Z., Shi, L., **Li, J.**, Dai, W., Lu, W., **Li, M.**, "The synergistic effects of GCPs and camera calibration models on UAV-SfM photogrammetry". In: *Drones* 9.5 (2025), p. 343

[5] Qiu, R., Dai, W., Wang, G., Luo, Z., **Li, M.**, "Evaluation of different deep learning methods for meteorological element forecasting". In: *IEEE Access* 12 (2024), pp. 81772–81782

[6] **Li, M.**, Dai, W., Fan, M., Qian, W., Yang, X., Tao, Y., Zhao, C., "Combining deep learning and hydrological analysis for identifying check dam systems from remote sensing images and DEMs in the Yellow River Basin". In: *International Journal of Environmental Research and Public Health* 20.5 (2023), p. 4636

[7] Chen, K., Wang, C., Lu, M., Dai, W., Fan, J., **Li, M.**, Lei, S., "Integrating topographic skeleton into deep learning for terrain reconstruction from GDEM and Google Earth image". In: *Remote Sensing* 15.18 (2023), p. 4490

Conference Presentations

[1] **Li, M.**, Meng, P., Fu, C., "When Association Is Not Enough: Explore Causal Mechanisms of Urban Mobility Resilience under Extreme Weather". In: The 18th International Conference on Location Based Services (LBS). Nanjing, China, 2026

[2] **Li, M.**, Weibel, R., Hu, Y., Fu, C., "Understanding the Resilience of Different Urban Mobility Systems to Extreme Weather Events: A Case Study of Hurricane Ida". In: Annual Meeting of the American Association of Geographers (AAG). San Francisco, USA, 2026

[3] **Li, M.**, Dai, W., "Reconstructing High-Resolution DEMs from 3D Terrain Features using Conditional Generative Adversarial Networks". In: Annual Meeting of the American Association of Geographers (AAG). Hawaii, USA, 2024

Professional Experience

- **Department of Geography, University at Buffalo** *Jan. 2026 - Present*
  Teaching Assistant Buffalo, NY
  ◦ GEO 481/506 Geographical Information Systems Lab
- **Chair of Applied Economics, ETH Zürich** *Oct. 2023 - Sep. 2024*
  Research Assistant Zurich, Switzerland
- **GeoSpatialX Lab, National University of Singapore** *Jun. 2023 - Nov. 2023*
  Research Assistant (remote) Singapore
- **SRK Consulting.** *Jan. 2022 - Feb. 2022*
  Cartography Intern Beijing, China
- **Esri (China) Information Technology Co., Ltd.** *Jun. 2021 - Sep. 2022*
  Trainee Technical Engineer Beijing, China

Service

Peer Reviewer

- *Journal of Planning Literature* 2026
- *International Journal of Applied Earth Observation and Geoinformation* 2026

Service

- Treasurer, Geography Graduate Student Association, University at Buffalo 2026–Present

Skills

- **Programming & ML:** Python, SQL, R, PyTorch, TensorFlow, Scikit-learn
- **Data Science Tools:** Pandas, NumPy,Stata, Jupyter, Git, Linux, PostgreSQL
- **Geospatial Tech:** Google Earth Engine (GEE), ArcGIS Pro, QGIS

Selected Awards

- Outstanding Graduates *Jun. 2023*
- First Class Scholarship *2020-2023*
- Silver Award in Esri-Cup College Students GIS Software Development Contest *Oct. 2021*

References

Available upon request.